INCH-POUND

MIL-PRF-1/1618C
4 September 1998
SUPERSEDING
MIL-E-1/1618B
24 March 1980

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON
TYPE 7256

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Pulsed, tunable frequency range: 9,000 to 9,160 MHz, integral magnet, 40-kW rated peak-power output.

ABSOLUTE RATINGS:

Parameter:	Ef	tk	tpc	Du	ib	epy	Pi	VSWR	Tuner torque	Altitude
Unit:	V	sec	μs	---	a	kV	W	---	inch-ounce	feet
Maximum	7.0	---	0.6	.0011	15.5	16	240	1.5	40	20,000
Minimum	---	120	---	---	12.5	---	---	---	---	---

PHYSICAL CHARACTERISTICS: See figure 1.

Mounting: Mounting flange	Input coupling: See figure 1
Mounting position: Any	Weight: 5 pounds (approximately)
Output coupling: 3/	Cooling: Air

TEST CONDITIONS:

Parameter: / Unit: / Tolerance:	Ef V ---	tk sec (max)	Du --- ---	tpc μs ±0.05	trv μs (min)	Ib mA dc ---	VSWR --- ----	Frequency MHz ---
Test condition 1:	6.3	120	---	---	---	---	---	---
Test condition 2:	0	---	.0010	0.5	0.08	14	1.1:1	F1
Test condition 3:	0	---	.0010	0.5	0.08	14	1.1:1	F1, F2, F3
Test condition 4:	0	---	.0010	0.5	0.08	14	1.1:1	F3

Test frequencies
F1 = 9,000 MHz
F2 = 9,080 MHz
F3 = 9,160 MHz

GENERAL:

Qualification - Required.

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TABLE I. Testing and inspection.

Inspection	Method	Test	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 1			1/				
Heater current	1301	1		If	0.90	1.10	A
Pressurizing	4003	---	40 lb_f /in^2 absolute (input and output)	---	---	---	---
Pulse voltage	4306	2		epy	13	15	kv
Power output	4250	3		Po	40	---	W
RF bandwidth	4308	3		BW	---	2.0/tpc	MHz
Minor lobes	4308	2		ratio	6	---	dB
Starting stability	4315	2	4/	MP	---	1.0	%
Conformance inspection, part 2			2/				
Low frequency vibration	1031		No voltages	---	---	---	---
Mechanical tuning range	4223	3	High frequency limit Low frequency limit	F F	9,160 ---	--- 9,000	MHz MHz
Tuner operation torque	4223		TA = 30°C (max)	Torque	---	10	inch-ounce
RF bandwidth	4308	3	VSWR = 1.5:1 Ib = 12.5 and 15.5 mA dc 5/	BW	---	2.2/tpc	MHz
Reverse tuning	---	3	6/	ΔF	---	200	kHz
Pulling factor	4310	2		---	---	---	---
Stability	4315	3	VSWR = 1.5:1 Ib = 15.5 mA dc 7/	MP	---	0.5	%
Conformance inspection, part 3							
Cycled life test	4551	3	VSWR = 1.5:1 (min) 11/	N	223	---	cycles
Life-test end points:	4551						
Power output	4250	3		Po	32	---	W
Pulse voltage	4306	2		epy	12	16	kV
Stability	4315	3	Ib = 15.5 mA dc 7/	MP	---	1.0	%
RF bandwidth	4308	2	Ib = 12.5 and 15.5 mA dc	BW	---	2.5/tpc	MHz
Minor lobes	4308	3		ratio	6	---	dB

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Test	Conditions	Symbol	Limits		Unit
					Min	Max	
Conformance inspection, part 3 - Continued							
High frequency vibration	1031		No voltages 12/	---	---	---	---
Shock	1041		8/ 12/	---	---	---	---
Air cooling	1143	2	TA = 50°C (max) Pi - Po = 150 W 9/ 12/ 14/	ΔT	---	TA+50	°C
Tuner operating torque	4223		TA = -55°C and +125° 12/	Torque	---	32	inch-ounce
Anode-cathode cap capacitance	---			C	4.0	8.0	pF
Mechanical fatigue	4223		10/ 12/	N	2,500	---	cycles
Temperature coefficient	4027	4	T(anode) = 70°C to 100°C 12/ 14/	ΔF/ΔT	---	0.25	MHz/°C
Barometric pressure reduced	1002		Pressure = 349 mmHg F = 9,160 MHz 12/	---	---	----	---
Low temperature operation	1047	4	Stability 7/ 12/	MP	---	2	%

1/ Unless otherwise specified, the acceptance level for all tests listed under conformance inspection, part 1, shall be 1.0 percent with inspection level of II.

2/ Unless otherwise specified, the acceptance level for all tests listed under conformance inspection, part 2, shall be 6.5 percent with inspection level of S3.

3/ The magnetron may be coupled directly to a RG-51/U waveguide in accordance with MIL-W-85, with a UG-52B/U chock flange in accordance with MIL-PRF-3922/25. However, for a minimum VSWR of this coupling, it is recommended that a specially designed choke flange be used which mates with the angular RG-51/U output of the magnetron to provide a straight section of RG-51/U waveguide through the coupling network.

4/ Immediately after the holding period, and after a 120-second preheat, the pulse voltage as specified in conformance inspection, part 1, shall be instantaneously applied and the magnetron missing pulse count shall not exceed 1 percent over the first minute of operation.

5/ The RF bandwidth shall be within the limits specified when a VSWR of 1.5 is introduced in the load at a distance no greater than 0.197 inches (0.5 meters) from the magnetron coupling flange, the phase being adjusted for the worst spectrum.

6/ When the direction of tuner drive is reversed at the frequency specified, the magnetron shall not continue to tune in the original direction by more than the specified frequency increment before reversing. Test to be performed at 9,000 MHz, and 9,080 MHz, and 9,160 MHz.

7/ Stability shall be measured with the VSWR adjusted to the phase which produces maximum instability. The missing pulses shall be counted during any 3 minute period of a test interval not to exceed 6 minutes.

8/ The tube shall be subjected to five shocks of peak amplitude duration in accordance with MIL-STD-202, method 213, test condition G. Following the shock test, the tube shall show no mechanical damage, and shall pass the power output and pulse voltage requirements of conformance inspection, part 1.

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TABLE I. Testing and inspection - Continued.

9/ An airflow, at standard atmospheric pressure, of 25 ft^3/min shall be directed at the cooling fins from an orifice of 1.1719 inches (2.977 mm) by 1.3594 inches (34.53 mm) located 0.250 inches (6.35 mm) from the cooling fins.

10/ The tuning shaft shall be continuously driven at a speed of approximately 650 r/min. The tube shall meet end-of-life requirements at completion of this test.

11/ Life test shall be conducted in accordance with the following cycle:

Test condition	Duration
1	2 minutes
3	4 hours
off	28 minutes

12/ This test need not be performed if last performed within 3 years and no related change has occurred in design, materials used, or production methods used in the manufacture of tube type 7256.

13/ The use of reclaimed materials shall be encouraged to the maximum extent possible.

14/ Anode temperature to be measured at this point.

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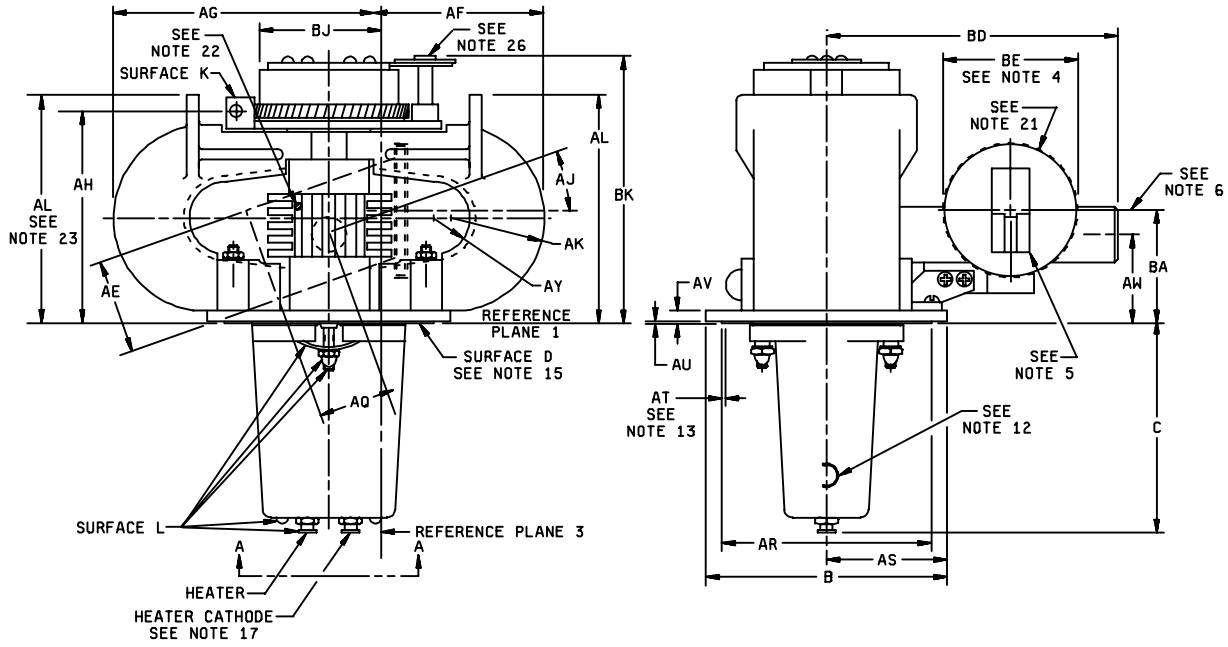


FIGURE 1. Outline drawing of electron tube type 7256

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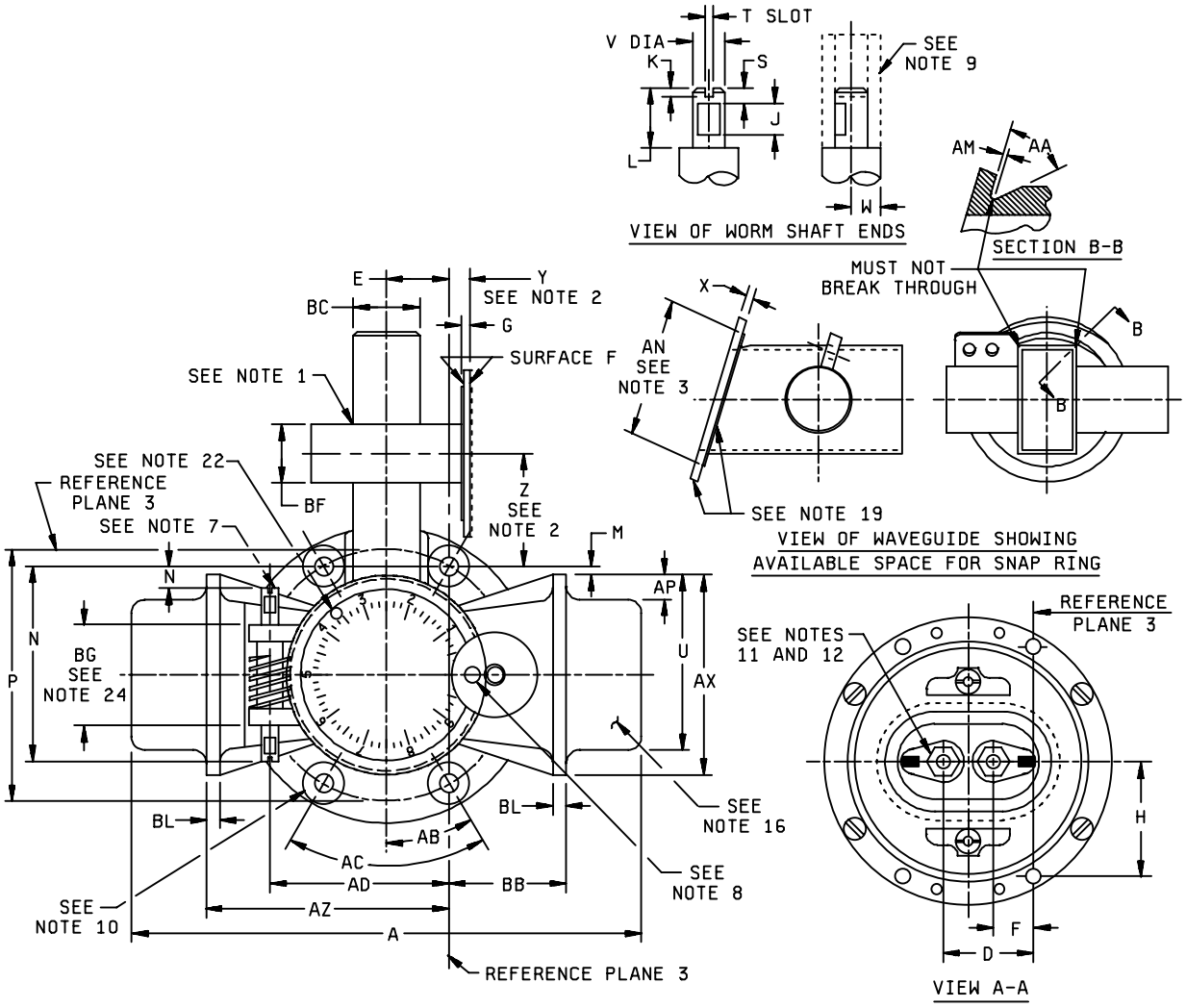


FIGURE 1. Outline drawing of electron tube type 7256 - Continued.

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Dimensions																
Ltr	Inches		Millimeters		Ltr	Inches		Millimeters		Ltr	Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max		Min	Max	Min	Max
Conformance inspection, part 1					Conformance inspection, part 2					BD	---	3.906	---	99.21
D	.969 NOM		24.61 NOM		A	---	5.937	---	150.80	BF	.625 Nom		15.88 Nom	
E	.719 NOM		18.26 NOM		B	3.219	3.281	81.76	83.34	BJ	---	1.700	---	43.18
F	.469 NOM		11.91 NOM		C	2.923	3.047	73.24	77.39	BK	---	4.200	---	106.70
H	1.245 NOM		31.62 NOM		J	.187	.203	4.75	5.16	BL	.1875 Nom		4.76 Nom	
K	.115	.135	2.92	3.43	L	.428	.448	10.67	11.38	Conformance inspection, part 3 12/				
M	.042	---	1.07	---	P	2.869	2.881	72.87	73.18	G	.125 Nom		3.18 Nom	
N	.230	.290	5.84	7.37	S	.140	.156	3.56	3.96	Q	1.438 Nom		36.53 Nom	
R	2.105	2.165	53.47	54.99	T	.040	.045	1.02	1.14	W	.151	.161	3.84	4.09
Y	.261	.301	6.63	7.65	U	---	2.187	---	55.55	X	.080	.090	2.03	2.29
Z	1.173	1.213	29.79	30.81	V	.1865	.1895	4.74	4.81	AA	45°	---	45°	---
AD	1.954	1.984	49.63	50.39	AC	59°48’	60°12’	---	---	AB	30°	---	30°	---
AH	2.834	2.874	71.98	72.00	AF	---	2.250	---	57.15	AE	1.250 Nom		31.75 Nom	
AM	.021	.031	0.53	0.79	AG	---	3.687	---	93.65	AJ	18°42’	---	---	---
AN	1.438	1.446	36.53	36.73	AL	---	3.281	---	83.34	AK	1.296 Nom		32.92 Nom	
AX	---	2.453	---	62.31	AP	.308	---	7.82	---	AQ	.930 Nom		23.62 Nom	
BA	1.542	1.582	39.17	40.18	AR	2.620	2.630	66.55	66.80	AU	.023	.039	0.58	0.99
BE	1.743	1.757	44.27	44.63	AS	1.625 Nom		41.28 Nom		AV	.156	---	3.96	---
BG	.990	1.010	25.15	25.65	AT	.057	.067	1.45	1.70	AW	1.224 Nom		31.09 Nom	
					AZ	---	2.781	---	70.64	AY	.484 Nom		12.29 Nom	
					BB	---	1.343	---	34.11	BC	.750 Nom		19.05 Nom	

FIGURE 1. Outline drawing of electron tube type 7256 - Continued.

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NOTES:

1. All solder joints on waveguide assembly shall be soldered to provide a hermetic seal at surface (F).
2. The ± .020 inch (0.51 mm) tolerance of .281 inch (7.14 mm) dimension includes angular as well as lateral deviations of the surface. These dimensions define relation of surface (F) to respective holes.
3. This diameter shall be concentric with the diameter of the flange within .005 inch (0.13 mm).
4. This diameter shall be concentric with opening in waveguide within .010 inch (0.25 mm).
5. The opening in the waveguide shall be enclosed by a dust cover when the tube is not in use.
6. Center line of waveguide opening.
7. To increase frequency, drive this end of worm shaft in counter clockwise direction. Complete frequency range covered in approximately 20 turns. Tuning mechanism shall operate over the entire mechanical range when subjected to a torque of 10 inch-ounces applied at the worm shaft. In equipment no less than 10 inch-ounces nor more than 2,500 inch-pounds shall be applied at drive shaft. The gear teeth and worm threads must be free from corrosion, paint, and other imperfections.
8. Number appearing here indicates number of complete revolutions of gear. With the geneva and gear set near 3 and zero respectively, the frequency of the tube is approximately 9,000 MHz.
9. It shall be possible for a sleeve .195 inch (4.95 mm) I. D. x .406 inch (10.31 mm) O. D. x 1.000 inch (25.40 mm) long to pass over ends of shaft to face of worm shaft housing.
10. Four .193 ± .003 inch (4.90 ± 0.08 mm) diameter holes. An area of .219 inch (5.56 mm) radius located from the center of each of the four holes to be free from finish.
11. The jack holes shall be within a radius of .023 inch (0.58 mm) of the specified location but shall be spaced .500 ± .010 inch (12.70 ± 0.25 mm) with respect to each other, the center lines of the jack holes shall be perpendicular to mounting plate within 3°.
12. 'C' indicates adjacent jack is the common heater-cathode connection.
13. With .062 inch (1.57 mm) seat resting on a plane surface the flatness of the seat shall be such that a .010 inch (0.25 mm) thickness gauge .125 inch (3.18 mm) wide shall not pass under.
14. All solder joints on base plate shall be soldered to provide a hermetic seal.
15. Any portion of the assembly extending below this surface shall be within a 1.109 inch (28.17 mm) radius on the true center of the plate.
16. All metal surfaces covered by finish except surfaces (D) and (F) drive and stop mechanism (K) surface around holes as specified in 10/, parts associated with insulator (L).
17. Locking type jack: See latest issue of MIL-PRF-1.
18. Hex head banana pin jack .594 inch (15.09 mm) long, hole .169 ± .005 inch (4.29 ± 0.13 mm).
19. Finish may be omitted from the edges and back of waveguide.
20. Warning: maintain minimum 2 inches (50.8 mm) clearance between this magnet and magnetic materials (magnets, steel tools, plates, etc.).

FIGURE 1. Outline drawing of electron tube type 7256 - Continued.

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NOTES - (Continued.):

21. Waveguide cover required for shipping purposes only.
22. Anode temperature to be measured at this point.
23. Includes top of worm gear housing.
24. This dimension applies from end of ears of worm shaft housing to a line .190 inch (4.83 mm) minimum from centerline of worm shaft toward reference plane 3.
25. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
26. Geneva cover retained by 6-32/.250 pan head screw and shakeproof washer.

FIGURE 1. Outline drawing of electron tube type 7256 - Continued.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Navy - AS, CG, OS, SH
Air Force - 11, 99

Preparing activity:
DLA - CC

(Project 5960-3495)

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